

Relay Catalysis for Selective Aerobic Oxidative Esterification of Primary Alcohols with Methanol

Yibo Yu,[#] Jie Lin,[#] Anni Qin,[#] Huanan Wang,[#] Jie Wang,[#] Weiyi Wang,[#] Guolin Wu,[#] Qian Zhang,[#] Hui Qian,^{*} and Shengming Ma^{*}



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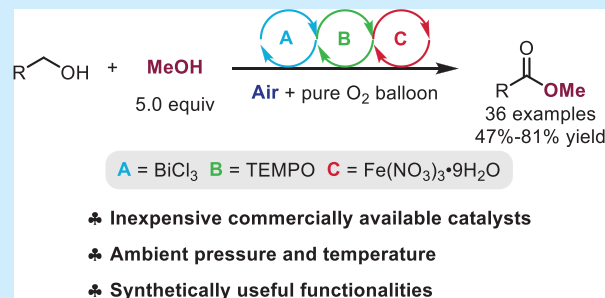
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ABSTRACT: Esters are bulk and fine chemicals and ubiquitous in polymers, bioactive compounds, and natural products. Their traditional synthetic approach is the esterification of carboxylic acids or their activated derivatives with alcohols. Herein, a bimetallic relay catalytic protocol was developed for the aerobic esterification of one alcohol in the presence of a slowly oxidizing alcohol, which has been identified as methanol. A concise synthesis of phlomic acid was executed to demonstrate the practicality and potential of this reaction.



Due to the global concern of natural resources and the environment,¹ more and more attention has been paid to enforce sustainable development of the whole society, thus, there is a great demand of the development of green production technology for the commodity chemicals and functional molecules.² Traditional oxidation reactions are an essential reaction for both the academic laboratory and industry, requiring at least stoichiometric amounts of oxidants, causing serious environmental burden. For this reason, catalytic aerobic oxidations, which use oxygen as the oxidant, have been actively pursued. The catalytic aerobic oxidations of primary alcohols, especially catalysis by transition metals such as Pd,^{3a,b} Au,^{3c,d} Ru,^{3e,f} Cu,^{3g,h} Fe,^{3i,j} etc., affording aldehydes or carboxylic acids have been well developed (Figure 1B).⁴ On the other hand, carboxylic acid esters are one of essential chemicals due to their widespread existence in bulk and fine chemicals, polymers, natural products, and drug molecules.^{5a} For some typical examples of methyl esters, see Figure 1A. The traditional approach is esterification of carboxylic acids and their derivatives, such as acyl halides, anhydrides, etc., with alcohols (Figure 1C).⁵ Dehydrogenative cross-coupling of alcohols to esters at high temperature is also a very useful strategy.⁶ The more ideal approach would be the aerobic oxidative esterification of the primary alcohol in the presence of another alcohol, but the accompanying challenges are (1) the selectivity issues referring to diversified oxidation by-products including aldehydes and carboxylic acids and (2) the selectivity issues forming four esters C1–C4 via both cross-esterification and self-esterification (Figure 1D).^{7–10} We proposed a concept of relay catalysis¹¹ with catalytic recipes A and B (Figure 1E). The catalytic recipe A could recognize the difference of the two primary alcohols, which oxidize one to aldehyde and leave the other one untouched. Then the

catalytic recipe B is responsible for the 1,2-addition of the in situ formed aldehyde with the unreacted alcohol to form the hemiacetal and its subsequent oxidation to the target ester. Thus, the goal is to identify the suitable relay catalysis and the slowly oxidizing alcohol. Here we report our successful development of such a bimetallic protocol of Fe and Bi for the highly selective aerobic oxidative cross esterification of alcohols with MeOH, which was identified as the very slowly oxidizing primary alcohol, with the help of a catalytic amount of TEMPO (Figure 1F).

Based on our hypothesis, the oxidative esterification of cetyl alcohol 1a with a series of different alcohols was conducted (Table 1). After tedious screening, BiCl₃ has been identified as the potential cocatalyst together with Fe(NO₃)₃·9H₂O and TEMPO for such a target transformation.¹² As reported, the reaction of 1a with benzyl alcohol 2a afforded the corresponding aldehydes 3a/8a⁷ and the self-esterification product 5a/10a^{9,10} (Table 1, entry 1); with *n*-BuOH, 51% of palmitaldehyde 3a and 32% of butyraldehyde 8b^{7h} were obtained with 11% of recovery of 1a. Disappointingly, due to the overlap of signals for esters 6ab, 5a, and 11ba, the specific yield of cross-esterification product 6ab was not able to be determined, and only the combined yield of three esters 6ab, 5a, and 11ba was available (Table 1, entry 2, for detail, see Supporting Information). Interestingly, with *n*-propanol, we

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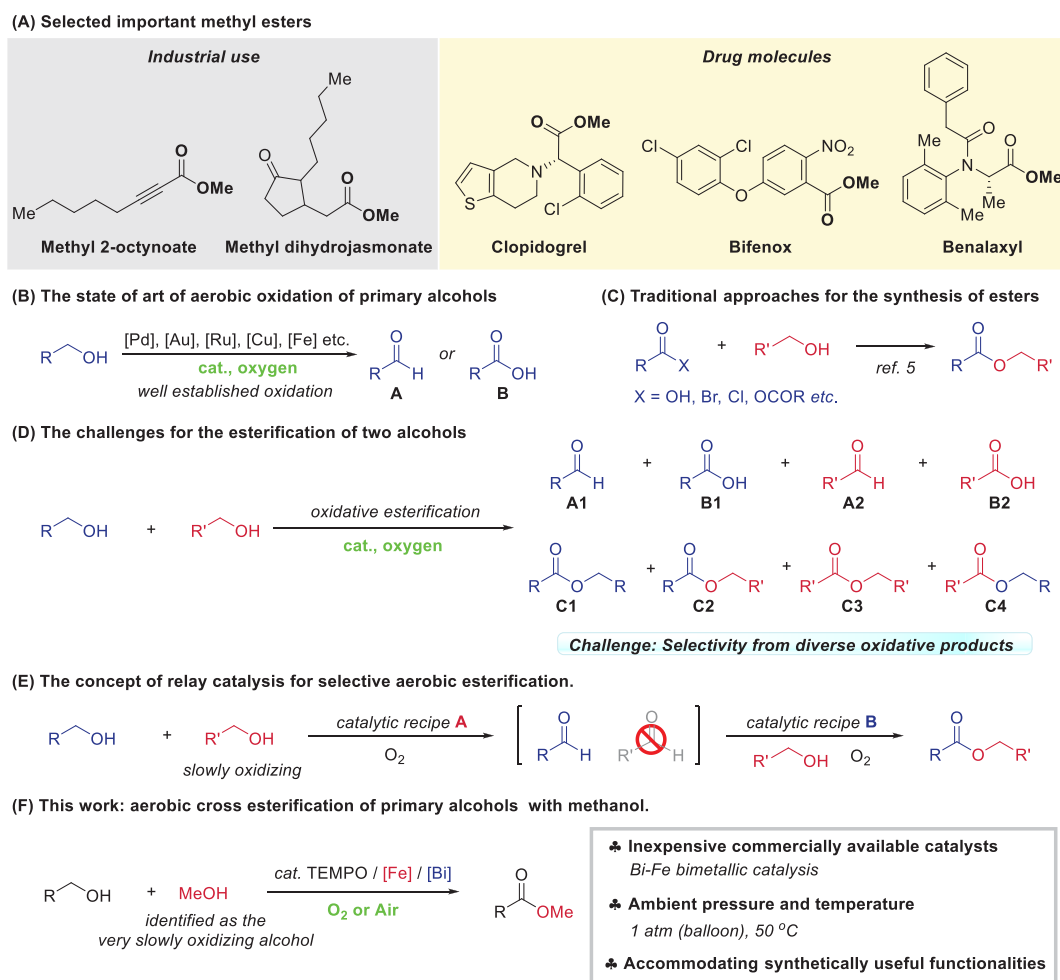
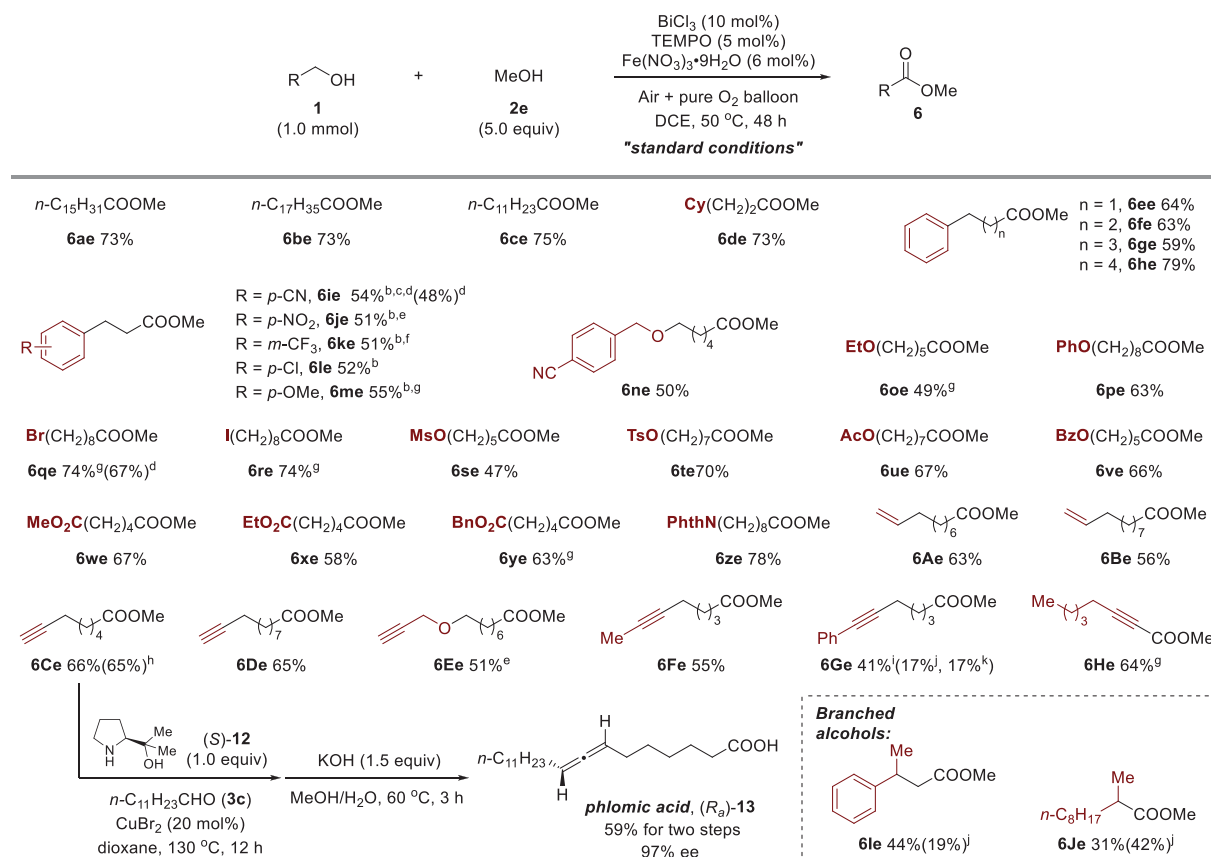


Figure 1. Approaches for the synthesis of esters. (A) Selected important methyl esters. (B) The state of the art of aerobic oxidation of primary alcohols. (C) Traditional approaches for the synthesis of esters. (D) The challenges for the esterification of two alcohols. (E) The concept of relay catalysis for selective aerobic esterification. (F) This work: aerobic cross esterification of primary alcohols with methanol.

Table 1. Identification of the Slowly Oxidizing Alcohols^a

1a (1.0 mmol)			3a	5a	6aa-af			
2a-f (5.0 equiv)			8a-f	10a-e (R ² = H)	11aa-ea (R ² = H)			
Entry		Recovery of 1a	NMR yield					
			3a	8	10	6	5a	11
1	BnOH (2a)	0	69%	86% (8a)	3% (10a)	0	10%	0
2	<i>n</i> -BuOH (2b)	11%	51%	32% (8b)	14% (10b)	19% ^b (6ab + 5a + 11ba)		
3	<i>n</i> -PrOH (2c)	9%	51%	14% (8c)	12% (10c)	8% (6ac)	12% ^c (5a + 11ca)	
4	EtOH (2d)	0	63%	12% (8d)	14% (10d)	14% (6ad)	9%	8% (11da)
5	MeOH (2e)	0	9%	ND ^d	9% (10e)	83% (6ae)	3%	1% (11ea)
6	<i>i</i> -PrOH (2f)	49%	24%	45% (8f)	-	0	3%	-

^aThe reaction was conducted with **1a** (1.0 mmol), $\text{R}'\text{CH}_2\text{OH}$ (5.0 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (6 mol %), nitroxyl (5 mol %), and Lewis acid (10 mol %) in 3 mL of solvent at T °C for 48 h with an O_2 balloon. The NMR yield and recovery were determined by ^1H NMR analysis using dibromomethane as the internal standard. ^bCombined yield of **6ab**, **5a**, and **11ba** based on alcohol **1a**. ^cCombined yield of **5a** and **11ca** based on alcohol **1a**. ^dND = not determined.

Scheme 1. Substrate Scope^a

^aThe reaction was conducted with **1** (1.0 mmol), MeOH (5.0 mmol), Fe(NO₃)₃·9H₂O (6 mol %), TEMPO (5 mol %), and BiCl₃ (10 mol %) in 3 mL of DCE at 50 °C for 48 h with an O₂ balloon. ^b4 mL of DCE were used. ^cThe reaction time was 60 h, 47% isolated yield. ^dNMR yield under standard conditions. ^e8 mol % each of Fe(NO₃)₃·9H₂O and TEMPO were used. ^f4 mmol of MeOH were used. ^g8 mol % of Fe(NO₃)₃·9H₂O was used. ^h5 mmol scale reaction. ⁱ10 mol % of TEMPO was used. ^jNMR yield of the aldehydes in parentheses determined by ¹H NMR analysis using dibromomethane as the internal standard. ^kNMR yield of the acetal determined by ¹H NMR analysis using dibromomethane as the internal standard.

surely observed the formation of the cross-esterification product **6ac** (Table 1, entry 3); with ethanol, the yield of the cross-esterification product **6ad** was improved to 14% (Table 1, entry 4). Surprisingly, the reaction with methanol afforded **6ae** in 83% yield highly selectively (Table 1, entry 5). As a comparison, 49% of alcohol **1a** was recovered and 45% of acetone **8f** was detected when the sterically more hindered *i*-propanol was applied instead of methanol (Table 1, entry 6). Thus, methanol was identified as the slowly oxidizing alcohol for the target transformation.^{13,14} For more detailed optimizations on the reaction of **1a** in the presence of methanol, see the Supporting Information (Table S2–8).

With the optimized reaction conditions in hand, we explored the generality of this method for the aerobic synthesis of methyl carboxylates from different primary alcohols (Scheme 1). Nonfunctionalized aliphatic alcohols and 3-cyclohexylpropanol performed well, with methyl esters **6ae–6de** being isolated in 73–75% yields. Aliphatic alcohols with different carbon chains containing phenyl also worked, providing **6ee–6he** in up to 79% yields. Additionally, 3-arylpropanols incorporating either electron-withdrawing or electron-donating groups, such as cyano, nitro, trifluoromethyl, chloro, and methoxy, were all smoothly oxidized to corresponding products **6ie–6me** in moderate yields with these functional groups untouched. It was worth noting that the yield of **6ie** was

increased from 48% to 54% with a slightly lower concentration. In addition, a variety of functional groups in alkyl alcohols, such as various ethers, highly reactive functionalities toward methanol, including bromo, iodo, MsO, TsO, amides, and even diverse esters, were well tolerated, affording **6ne–6ze** in moderate to good yields.

The C=C bond was readily accommodated, providing **6Ae** and **6Be** smoothly. In addition, both terminal and internal C–C triple bonds were compatible, providing the corresponding methyl esters **6Ce–6He** in satisfactory yields. The corresponding aldehyde and acetal were also formed in 17% NMR yield, respectively, under standard conditions, accounting for the lower yield for **6Ge**. Notably, methyl oct-2-ynoate **6He**, which is commonly used as flavoring agents to prepare cucumber, banana, strawberry, peach, etc., was obtained easily from the corresponding alcohol **1H** under the standard conditions in 64% yield. The reactions of branched alcohols **1I** and **1J** with methanol were much slower, affording the methyl esters **6Ie** and **6Je** in 44% and 33% yield, respectively, together with the related aldehydes **3I** and **3J**, which could be attributed to the increased steric hindrance. By applying the present protocol following enantioselective allenation of alkynes reaction (EATA),^{12d,15} a three-step concise synthesis of phlomic acid isolated from *Phlomis* in 1997¹⁶ has been realized successfully. First, a 5 mmol scale reaction afforded methyl hept-6-ynoate

6Ce in 65% isolated yield, which underwent EATA reaction with dodecaldehyde **3c** in the presence of (*S*)-dimethylprolinol (*S*)-**12**. Pholmic acid (*R_a*)-**13** was obtained by hydrolysis in the presence of KOH in 59% total yield with 97% ee.

In conclusion, we have successfully developed a BiCl₃ and iron nitrate cocatalyzed selective oxidative esterification of primary alcohols in the presence of methanol with molecular oxygen as an environmentally benign oxidant with the help of TEMPO. A wide range of alcohols bearing various highly reactive functional groups, such as halogens, ethers, esters, amide, C=C, C≡C, etc., may be converted to the corresponding methyl esters in moderate to high yields. A highly concise three-step enantioselective synthesis of pholmic acid has been achieved by using the current protocol as the key step. Further studies are being actively pursued in our laboratory.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its [Supporting Information](#)

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.orglett.4c01059>.

Optimizations, experimental procedures, and NMR spectra for obtained compounds (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Hui Qian – Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, P. R. China; orcid.org/0000-0002-5606-9213; Email: qian_hui@fudan.edu.cn

Shengming Ma – Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, P. R. China; State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, Shanghai 200032, P. R. China; orcid.org/0000-0002-2866-2431; Email: masm@sioc.ac.cn

Authors

Yibo Yu – Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, P. R. China

Jie Lin – Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, P. R. China

Anni Qin – Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, P. R. China

Huanan Wang – Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, P. R. China

Jie Wang – Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, P. R. China

Weiyi Wang – Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, P. R. China

Guolin Wu – Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, P. R. China

Qian Zhang – Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, P. R. China

Complete contact information is available at:

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Author Contributions

[#]These authors contributed equally.

Notes

The authors declare no competing financial interest.

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